

Hi-MO X10

Anti-Dust PRO

LR8-66HVDF 640~670M

- Advanced Nano-Ceramic Coating: Hydrophobic, anti-static properties reduce dust adhesion and maintain high optical transmittance.
- Edge-to-edge glass-to-frame design prevents lower-edge dust accumulation and associated soiling losses.
- Innovative anti-dust structural design delivers an average energy yield increase of $\geq 2\%$.
- Optimized O&M, with reduced cleaning cycles up to 40%.
- Back-contact cell technology integrates anti-shading functionality to reduce shading losses, localized heating, and hotspot risk.

15

15-year Warranty for
Materials and Processing

30

30-year Warranty for Extra
Linear Power Output

Complete System and Product Certifications

IEC 61215, IEC 61730, UL 61730

ISO9001:2015: ISO Quality Management System

ISO14001: 2015: ISO Environment Management System

ISO45001: 2018: Occupational Health and Safety

IEC62941: Guideline for module design qualification and type approval

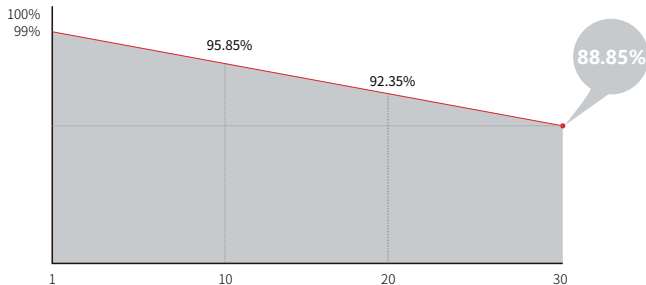
LONGI



24.8%
MAX MODULE
EFFICIENCY**0~3%**
POWER
TOLERANCE**<1%**
FIRST YEAR
POWER DEGRADATION**0.35%**
YEAR 2-30
POWER DEGRADATION**BC-CELL**
LOWER OPERATING
TEMPERATURE

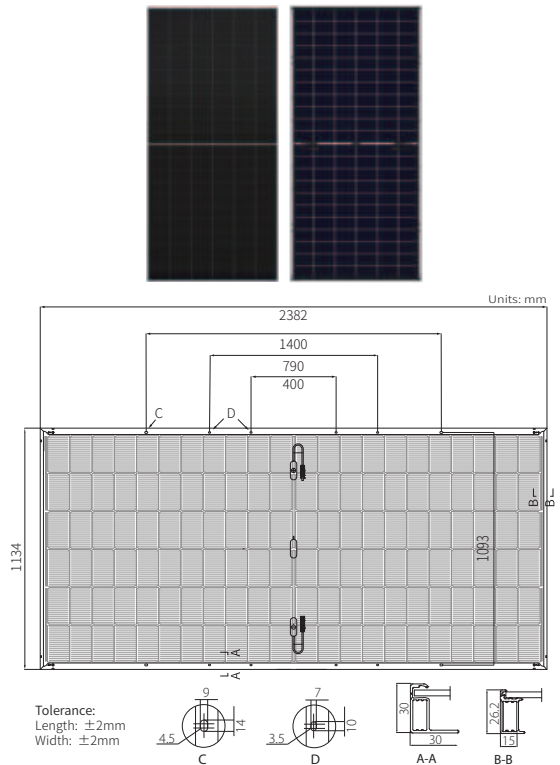
Additional Value

30-Year Power Warranty



Mechanical Parameters

Cell Orientation	132 (6×22)
Junction Box	IP68, three diodes
Output Cable	4mm ² , +400, -200mm/±1400mm length can be customized
Glass	Dual glass, 2.0+2.0mm heat strengthened glass
Frame	Anodized aluminum alloy frame
Weight	32.6kg
Dimension	2382×1134×30mm
Packaging	35pcs per pallet / 140pcs per 20' GP / 700pcs per 40' HC



Electrical Characteristics

STC : AM1.5 1000W/m² 25°CNOCT : AM1.5 800W/m² 20°C 1m/s

Test uncertainty for Pmax: ±3%

Module Type	LR8-66HVDF-640M		LR8-66HVDF-645M		LR8-66HVDF-650M		LR8-66HVDF-655M		LR8-66HVDF-660M		LR8-66HVDF-665M		LR8-66HVDF-670M	
Testing Condition	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT
Maximum Power (Pmax/W)	640	487.5	645	491.5	650	495.3	655	498.6	660	503.0	665	506.7	670	510.5
Open Circuit Voltage (Voc/V)	49.52	47.06	49.62	47.16	49.72	47.25	49.82	47.35	49.92	47.44	50.02	47.54	50.12	47.63
Short Circuit Current (Isc/A)	16.38	13.16	16.46	13.22	16.54	13.28	16.62	13.35	16.70	13.41	16.78	13.48	16.86	13.54
Voltage at Maximum Power (Vmp/V)	40.78	38.76	40.88	38.85	40.98	38.95	41.08	39.04	41.18	39.14	41.28	39.23	41.38	39.33
Current at Maximum Power (Imp/A)	15.69	12.58	15.78	12.65	15.86	12.72	15.94	12.78	16.03	12.85	16.11	12.92	16.19	12.98
Module Efficiency(%)	23.7		23.9		24.1		24.2		24.4		24.6		24.8	

Electrical characteristics with different rear side power gain (reference to 645W front)

Pmax /W	Voc/V	Isc /A	Vmp/V	Imp /A	Pmax gain
677	49.62	17.28	40.88	16.57	5%
710	49.62	18.11	40.88	17.36	10%
744	49.62	18.93	40.98	18.15	15%
776	49.72	19.75	40.98	18.94	20%
808	49.72	20.58	40.98	19.73	25%

Operating Parameters

Operational Temperature	-40°C ~ +85°C
Power Output Tolerance	0 ~ 3%
Maximum System Voltage	DC1500V (IEC/UL)
Maximum Series Fuse Rating	35A
Nominal Operating Cell Temperature	45±2°C
Protection Class	Class II
Bifaciality	75±5%
Fire Rating	UL type 29 IEC Class C

Mechanical Loading

Front Side Maximum Static Loading	5400Pa
Rear Side Maximum Static Loading	2400Pa
Hailstone Test	25mm Hailstone at the speed of 23m/s

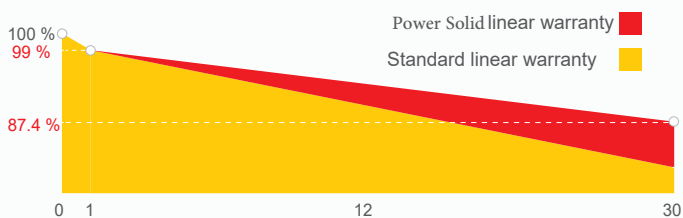
Temperature Ratings (STC)

Temperature Coefficient of Isc	+0.050%/°C
Temperature Coefficient of Voc	-0.200%/°C
Temperature Coefficient of Pmax	-0.260%/°C

PS620W#PVBN

BIFACIAL MODULE

Industry-leading Warranty based on nominal power



Features



Extended wind and snow load tests

Module certified to withstand extreme wind (2400 Pascal) and snow loads (5400 Pascal) *



Hot 2.0 Technology

The N-type module with Hot 2.0 technology has better reliability and lower LID/LETID.



Excellent weak light performance

More power output in weak light condition, such as cloudy, morning and sunset



SMBB Technology

Better light trapping and current collection to improve module power output and reliability.



Lower LCOE

Higher bifaciality, higher power output and lower BOS cost



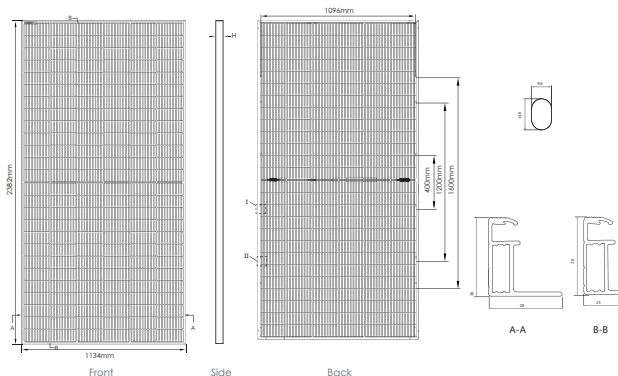
PID Resistance

Excellent Anti-PID performance guarantee via optimized mass-production process and materials control.

General Features

MECHANICAL SPECIFICATIONS

Cell Type	N type Mono-crystalline
Cell Dimensions	191.6*182mm
Cell Arrangement	144 (6*12)
Weight	33.5KG
Module Dimensions	2382*1134*30mm
Cables	4.0mm2(+): 300mm , (-): 300mm or Customized Length
Front Glass	2.0mm, Anti-Reflection Coating
Back Glass	2.0mm, Heat Strengthened Glass
No. of Bypass Diodes	3/6
Packing Configuration	36pcs/pallet, 720pcs/40hq
Frame	Anodized Aluminium Alloy
Junction Box	IP68



ELECTRICAL SPECIFICATIONS

Module Type	PS620W#PVBN	
Testing Condition	STC	NMOT
Rated output (Pmp/Wp)	620	466
Maximum Power Voltage(Vmpp/V)	45.90	42.57
Maximum Power Current(Imp/A)	13.51	10.95
Open Circuit Voltage(Voc/V)	55.56	52.80
Short Circuit Current(Isc/A)	14.19	11.45
Module efficiency(%)	23.0%	
Power Tolerance (W)	0~+5	

STC: Irradiance 1000W/m2, Cell Temperature 25°C, Air Mass AM1.5 NMOT: Irradiance at 800W/m2, Ambient Temperature 20°C, Air Mass AM1.5, Wind Speed 1m/s

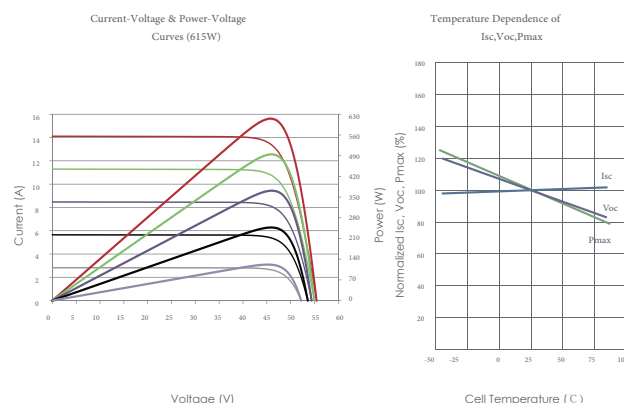
MAXIMUM RATINGS

Maximum System Voltage	1500V DC(IEC)
Operating Temperature	-40°C ~ +85°C
Maximum Series Fuse	30A
Static Loading	Snow Loading: 5400Pa/ Wind Loading: 2400Pa
Conductivity at Ground	≤0.1Ω
Safety Class	II
Resistance	≥100MΩ
Refer. Bifacial Factor	80% ± 5%

TEMPERATURE CHARACTERISTICS

NMOT Temperature	45°C ± 2°C
Temperature Coefficient (Pmax)	-0.30%/°C
Temperature Coefficient (Voc)	-0.25%/°C
Temperature Coefficient (Isc)	0.046%/°C

CURVE & TEMPERATURE DEPENDENCE



Single Phase Hybrid Inverter

SUN-3K-SG06LP1-24-EU-CM1 SUN-3K-SG06LP1-EU-CM1
SUN-3.6/4/4.6/5/6K-SG06LP1-EU-CM2



Intuitive

- Colorful touch LCD, IP65 protection degree



Intelligent

- 6 time periods for battery charging/discharging



Flexible

- AC couple to retrofit existing solar system
- Max. 16 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel



Reliable

- Max. charging/discharging current of 140A
- Support storing energy from diesel generator

Model	SUN-3K-SG06 LP1-24-EU-CM1	SUN-3K-SG06 LP1-EU-CM1	SUN-3.6K-SG06 LP1-EU-CM2	SUN-4K-SG06 LP1-EU-CM2	SUN-4.6K-SG06 LP1-EU-CM2	SUN-5K-SG06 LP1-EU-CM2	SUN-6K-SG06 LP1-EU-CM2
Battery Input Data							
Battery Type	Lead-acid or Lithium-ion						
Battery Voltage Range (V)	20-30	40-60					
Max. Charging/Discharging Current (A)	140	70	90	95	120	120	135
Charging Strategy for Li-ion Battery	Self-adaption to BMS						
Number of Battery Input	1						
PV String Input Data							
Max. PV access power (W)	6000	6000	7200	8000	9200	10000	12000
Max. PV Input Power (W)	4800	4800	5760	6400	7360	8000	9600
Max. PV Input Voltage (V)	500						
Start-up Voltage (V)	125						
MPPT Voltage Range (V)	150-425						
Rated PV Input Voltage (V)	370						
Max. Operating PV Input Current (A)	18		18+18				
Max. Input Short-Circuit Current (A)	27		27+27				
No. of MPP Trackers/ No. of Strings MPP Tracker	1/1		2/1+1				
AC Input/Output Data							
Rated AC Input/Output Active Power (W)	3000	3000	3600	4000	4600	5000	6000
Max. AC Input/Output Apparent Power (VA)	3300	3300	3960	4400	4600	5500	6600
Rated AC Input/Output Current (A)	13.7/13.1	13.7/13.1	16.4/15.7	18.2/17.4	21/20	22.8/21.8	27.3/26.1
Max. AC Input/Output Current (A)	15/14.4	15/14.4	18/17.3	20/19.2	21/20	25.0/24.0	30/28.7
Max. Continuous AC Passthrough (grid to load) (A)	50						
Peak Power (off-grid) (W)	2 times of rated power, 10s						
Power Factor Adjustment Range	0.8 leading to 0.8 lagging						
Rated input/output voltage	220/230V 0.85Un-1.1Un L+N+PE						
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65						
Total Current Harmonic Distortion THDi	<3% (of nominal power)						
DC Injection Current	<0.5% In						
Efficiency							
Max. Efficiency	97.6%						
Euro Efficiency	96.5%						
MPPT Efficiency	>99%						
Equipment Protection							
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI)(optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection						
Surge Protection Level	TYPE II(DC), TYPE II(AC)						
Interface							
LCD/LED Display	LCD						
Communication Interface	WIFI, RS485, CAN						
General Data							
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating						
Permissible Ambient Humidity	0-100%						
Permissible Altitude	3000m						
Noise (dB)	<45						
Ingress Protection(IP) Rating	IP 65						
Inverter Topology	Non-Isolated (solar), Isolated (battery)						
Over Voltage Category	OVC II(DC), OVC III(AC)						
Cabinet Size (WxHxD mm)	310×460×218 (Excluding Connectors and Brackets)						
Weight (kg)	15.6						
Type of Cooling	Natural Cooling						
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.						
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, VDE-AR-N 4105						
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2						

Single Phase Hybrid Inverter

SUN-7/7.6/8/10K-SG05LP1-EU-SM2



Intuitive

- Colorful touch LCD, IP65 protection degree



Intelligent

- 6 time periods for battery charging/discharging



Flexible

- AC couple to retrofit existing solar system
- Max. 16 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel



Reliable

- Max. charging/discharging current of 210A
- Support storing energy from diesel generator

Model	SUN-7K-SG05LP1 -EU-SM2	SUN-7.6K-SG05LP1 -EU-SM2	SUN-8K-SG05LP1 -EU-SM2	SUN-10K-SG05LP1 -EU-SM2
Battery Input Data				
Battery Type	Lead-acid or Lithium-ion			
Battery Voltage Range (V)	40-60			
Max. Charging/Discharging Current (A)	175	190	190	210
Charging Strategy for Li-ion Battery	Self-adaption to BMS			
Number of Battery Input	1			
PV String Input Data				
Max. PV access power (W)	14000	15200	16000	20000
Max. PV Input Power (W)	11200	12160	12800	16000
Max. PV Input Voltage (V)	500			
Start-up Voltage (V)	125			
MPPT Voltage Range (V)	150-425			
Rated PV Input Voltage (V)	370			
Max. Operating PV Input Current (A)	26+26			
Max. Input Short-Circuit Current (A)	34+34			
No. of MPP Trackers/ No. of Strings MPP Tracker	2/2+2			
AC Input/Output Data				
Rated AC Input/Output Active Power (W)	7000	7600	8000	10000
Max. AC Input/Output Apparent Power (VA)	7700	8360	8800	11000
Rated AC Input/Output Current (A)	31.9/30.5	34.6/33.1	36.4/34.8	45.5/43.5
Max. AC Input/Output Current (A)	35/33.5	38/36.4	40/38.3	50/47.9
Max. Continuous AC Passthrough (grid to load) (A)	50			
Peak Power (off-grid) (W)	2 times of rated power, 10s			
Power Factor Adjustment Range	0.8 leading to 0.8 lagging			
Rated input/output voltage	220/230V 0.85Un-1.1Un L+N+PE			
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65			
Total Current Harmonic Distortion THDi	<3% (of nominal power)			
DC Injection Current	<0.5% In			
Efficiency				
Max. Efficiency	97.6%			
Euro Efficiency	96.5%			
MPPT Efficiency	>99%			
Equipment Protection				
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection			
Surge Protection Level	TYPE II(DC), TYPE II(AC)			
Interface				
LCD/LED Display	LCD			
Communication Interface	WIFI,RS485,CAN			
General Data				
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating			
Permissible Ambient Humidity	0-100%			
Permissible Altitude	2000m			
Noise (dB)	<45			
Ingress Protection(IP) Rating	IP 65			
Inverter Topology	Non-Isolated			
Over Voltage Category	OVC II(DC), OVC III(AC)			
Cabinet Size (WxHxD mm)	366×589.5×237 (Excluding Connectors and Brackets)			
Weight (kg)	26.8			
Type of Cooling	Intelligent Air Cooling			
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.			
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, VDE-AR-N 4105			
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2			

Single Phase Hybrid Inverter

SUN-7.6/8K-SG02LP1-EU-AM2
SUN-10/12K-SG02LP1-EU-AM3



Intuitive

- Colorful touch LCD, IP65 protection degree



Intelligent

- 6 time periods for battery charging/discharging



Flexible

- AC couple to retrofit existing solar system
- Max. 16 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel



Reliable

- Max. charging/discharging current of 250A
- Support storing energy from diesel generator

Model	SUN-7.6K-SG02 LP1-EU-AM2	SUN-8K-SG02 LP1-EU-AM2	SUN-10K-SG02 LP1-EU-AM3	SUN-12K-SG02 LP1-EU-AM3
Battery Input Data				
Battery Type	Lead-acid or Lithium-ion			
Battery Voltage Range (V)	40-60			
Max. Charging/Discharging Current (A)	190	190	220	250
Charging Strategy for Li-ion Battery	Self-adaption to BMS			
Number of Battery Input	1			
PV String Input Data				
Max. PV access power (W)	15200	16000	20000	24000
Max. PV Input Power (W)	12160	12800	16000	19200
Max. PV Input Voltage (V)	500			
Start-up Voltage (V)	125			
MPPT Voltage Range (V)	150-425			
Rated PV Input Voltage (V)	370			
Max. Operating PV Input Current (A)	26+26		26+26+26	
Max. Input Short-Circuit Current (A)	44+44		44+44+44	
No. of MPP Trackers/ No. of Strings MPP Tracker	2/2+2		3/2+2+2	
AC Input/Output Data				
Rated AC Input/Output Active Power (W)	7600	8000	10000	12000
Max. AC Input/Output Apparent Power (VA)	8360	8800	11000	13200
Rated AC Input/Output Current (A)	34.6/33.1	36.4/34.8	45.5/43.5	54.6/52.2
Max. AC Input/Output Current (A)	38/36.4	40/38.3	50/47.9	60/57.4
Max. Continuous AC Passthrough (grid to load) (A)	50		60	
Peak Power (off-grid) (W)	2 times of rated power, 10s			
Power Factor Adjustment Range	0.8 leading to 0.8 lagging			
Rated input/output voltage	220/230V 0.85Un-1.1Un L+N+PE			
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65			
Total Current Harmonic Distortion THDi	<3% (of nominal power)			
DC Injection Current	<0.5 In			
Efficiency				
Max. Efficiency	97.6%			
Euro Efficiency	96.5%			
MPPT Efficiency	>99%			
Equipment Protection				
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection			
Surge Protection Level	TYPE II(DC), TYPE II(AC)			
Interface				
LCD/LED Display	LCD			
Communication Interface	WIFI, RS485, CAN			
General Data				
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating			
Permissible Ambient Humidity	0-100%			
Permissible Altitude	2000m			
Noise (dB)	<45			
Ingress Protection(IP) Rating	IP 65			
Inverter Topology	Non-Isolated (solar), Isolated (battery)			
Over Voltage Category	OVC II(DC), OVC III(AC)			
Cabinet Size (WxHxD mm)	420×670×233 (Excluding Connectors and Brackets)			
Weight (kg)	35.6			
Type of Cooling	Intelligent Air Cooling			
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.			
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, VDE-AR-N 4105			
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2			

Single Phase Hybrid Inverter

SUN-12/14/16K-SG01LP1-EU



Intuitive

- Colorful touch LCD, IP65 protection degree



Intelligent

- 6 time periods for battery charging/discharging



Flexible

- AC couple to retrofit existing solar system
- Max. 16 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel



Reliable

- Max. charging/discharging current of 290A
- Support storing energy from diesel generator

Model	SUN-12K-SG01LP1-EU		SUN-14K-SG01LP1-EU		SUN-16K-SG01LP1-EU	
Battery Input Data						
Battery Type	Lead-acid or Lithium-ion					
Battery Voltage Range (V)	40-60					
Max. Charging/Discharging Current (A)	220	250		290		
Charging Strategy for Li-ion Battery	Self-adaption to BMS					
Number of Battery Input	2					
PV String Input Data						
Max. PV access power (W)	24000	28000		32000		
Max. PV Input Power (W)	19200	22400		25600		
Max. PV Input Voltage (V)	500					
Start-up Voltage (V)	125					
MPPT Voltage Range (V)	150-425					
Rated PV Input Voltage (V)	370					
Max. Operating PV Input Current (A)	32+32+32					
Max. Input Short-Circuit Current (A)	60+60+60					
No. of MPP Trackers/ No. of Strings MPP Tracker	3/2+2+2					
AC Input/Output Data						
Rated AC Input/Output Active Power (W)	12000	14000		16000		
Max. AC Input/Output Apparent Power (VA)	13200	15400		17600		
Rated AC Input/Output Current (A)	54.5/52.2	63.6/60.9		72.7/69.6		
Max. AC Input/Output Current (A)	60/57.4	70/67		80/76.5		
Max. Continuous AC Passthrough (grid to load) (A)	100					
Peak Power (off-grid) (W)	2 times of rated power, 10s					
Power Factor Adjustment Range	0.8 leading to 0.8 lagging					
Rated input/output voltage	220/230V 0.85Un-1.1Un L+N+PE					
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65					
Total Current Harmonic Distortion THDi	<3% (of nominal power)					
DC Injection Current	<0.5% In					
Efficiency						
Max. Efficiency	97.6%					
Euro Efficiency	96.5%					
MPPT Efficiency	>99%					
Equipment Protection						
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI) (optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection					
Surge Protection Level	TYPE II(DC), TYPE II(AC)					
Interface						
LCD/LED Display	LCD					
Communication Interface	WIFI, RS485, CAN					
General Data						
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating					
Permissible Ambient Humidity	0-100%					
Permissible Altitude	2000m					
Noise (dB)	<45					
Ingress Protection(IP) Rating	IP 65					
Inverter Topology	Non-Isolated (solar), Isolated (battery)					
Over Voltage Category	OVC II(DC), OVC III(AC)					
Cabinet Size (WxHxD mm)	464×763×282 (Excluding Connectors and Brackets)					
Weight (kg)	52					
Type of Cooling	Intelligent Air Cooling					
Warranty	5 Years/10 Years the Warranty Period Depends the Final Installation Site of Inverter, More Info Please Refer to Warranty Policy					
Grid Regulation	IEC 61727,IEC 62116,NRS 097					
Safety EMC/Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2					

Three Phase Hybrid Inverter

SUN-3/4/5/6/8/10/12K-SG05LP3-EU-SM2



Safe

- 48V low voltage battery, transformer isolation design



Intelligent

- 6 time periods for battery charging/discharging



Flexible

- AC couple to retrofit existing solar system
- Max. 10 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel



Reliable

- Max. charging/discharging current of 240A
- Support storing energy from diesel generator
- 100% unbalanced output, max. output up to 50% rated power for each phase

Model	SUN-3K-SG05 LP3-EU-SM2	SUN-4K-SG05 LP3-EU-SM2	SUN-5K-SG05 LP3-EU-SM2	SUN-6K-SG05 LP3-EU-SM2	SUN-8K-SG05 LP3-EU-SM2	SUN-10K-SG05 LP3-EU-SM2	SUN-12K-SG05 LP3-EU-SM2
Battery Input Data							
Battery Type	Lead-acid or Lithium-ion						
Battery Voltage Range (V)	40-60						
Max. Charging/Discharging Current (A)	70	95	120	135	190	210	240
Charging Strategy for Li-ion Battery	Self-adaption to BMS						
Number of Battery Input	1						
PV String Input Data							
Max. PV access power (W)	6000	8000	10000	12000	16000	20000	24000
Max. PV Input Power (W)	4800	6400	8000	9600	12800	16000	19200
Max. PV Input Voltage (V)	800						
Start-up Voltage (V)	160						
MPPT Voltage Range (V)	200-650						
Rated PV Input Voltage (V)	550						
Max. Operating PV Input Current (A)	20+20					26+26	
Max. Input Short-Circuit Current (A)	30+30					39+39	
No. of MPP Trackers/ No. of Strings MPP Tracker	2/1+1					2/2+2	
AC Input/Output Data							
Rated AC Input/Output Active Power (W)	3000	4000	5000	6000	8000	10000	12000
Max. AC Input/Output Apparent Power (VA)	3300	4400	5500	6600	8800	11000	13200
Rated AC Input/Output Current (A)	4.6/4.4	6.1/5.8	7.6/7.3	9.1/8.7	12.2/11.6	15.2/14.5	18.2/17.4
Max. AC Input/Output Current (A)	5/4.8	6.7/6.4	8.4/8	10/9.6	13.4/12.8	16.7/16	20/19.2
Max. Continuous AC Passthrough (grid to load) (A)	45						
Peak Power (off-grid) (W)	2 times of rated power, 10s						
Power Factor Adjustment Range	0.8 leading to 0.8 lagging						
Rated input/output voltage	220/380V,230/400V 0.85Un-1.1Un 3L+N+PE						
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65						
Total Current Harmonic Distortion THDi	<3% (of nominal power)						
DC Injection Current	<0.5% In						
Efficiency							
Max. Efficiency	97.6%						
Euro Efficiency	97.0%						
MPPT Efficiency	>99%						
Equipment Protection							
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI)(optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection						
Surge Protection Level	TYPE II(DC), TYPE II(AC)						
Interface							
LCD/LED Display	LCD						
Communication Interface	WIFI,RS485,CAN						
General Data							
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating						
Permissible Ambient Humidity	0-100%						
Permissible Altitude	3000m						
Noise (dB)	≤55						
Ingress Protection(IP) Rating	IP 65						
Inverter Topology	Non-Isolated (solar), Isolated (battery)						
Over Voltage Category	OVC II(DC), OVC III(AC)						
Cabinet Size (WxHxD mm)	386×660×250 (Excluding Connectors and Brackets)						
Weight (kg)	35.2						
Type of Cooling	Intelligent Air Cooling						
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.						
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, G98, VDE-AR-N 4105						
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2						

Three Phase Hybrid Inverter

SUN-14/15/16/18/20K-SG05LP3-EU-SM2



Safe

- 48V low voltage battery, transformer isolation design



Intelligent

- 6 time periods for battery charging/discharging



Flexible

- AC couple to retrofit existing solar system
- Max. 10 pcs parallel for on-grid and off-grid operation; support multiple batteries parallel



Reliable

- Max. charging/discharging current of 350A
- Support storing energy from diesel generator
- 100% unbalanced output, max. output up to 50% rated power for each phase

Model	SUN-14K-SG05 LP3-EU-SM2	SUN-15K-SG05 LP3-EU-SM2	SUN-16K-SG05 LP3-EU-SM2	SUN-18K-SG05 LP3-EU-SM2	SUN-20K-SG05 LP3-EU-SM2
Battery Input Data					
Battery Type	Lead-acid or Lithium-ion				
Battery Voltage Range (V)	40-60				
Max. Charging/Discharging Current (A)	260	280	300	330	350
Charging Strategy for Li-ion Battery	Self-adaption to BMS				
Number of Battery Input	2				
PV String Input Data					
Max. PV access power (W)	28000	30000	32000	36000	40000
Max. PV Input Power (W)	22400	24000	25600	28800	32000
Max. PV Input Voltage (V)	800				
Start-up Voltage (V)	160				
MPPT Voltage Range (V)	160-650				
Rated PV Input Voltage (V)	550				
Max. Operating PV Input Current (A)	36+36				
Max. Input Short-Circuit Current (A)	54+54				
No. of MPP Trackers/ No. of Strings MPP Tracker	2/2+2				
AC Input/Output Data					
Rated AC Input/Output Active Power (W)	14000	15000	16000	18000	20000
Max. AC Input/Output Apparent Power (VA)	15400	16500	17600	19800	22000
Rated AC Input/Output Current (A)	21.3/20.3	22.8/21.8	24.3/23.2	27.3/26.1	30.4/29
Max. AC Input/Output Current (A)	23.4/22.4	25/24	26.7/25.6	30/28.7	33.4/31.9
Max. Continuous AC Passthrough (grid to load) (A)	70				
Peak Power (off-grid) (W)	2 times of rated power, 10s				
Power Factor Adjustment Range	0.8 leading to 0.8 lagging				
Rated input/output voltage	220/380V,230/400V 0.85Un-1.1Un 3L+N+PE				
Rated Input/Output Grid Frequency/Range(Hz)	50/45-55, 60/55-65				
Total Current Harmonic Distortion THDi	<3% (of nominal power)				
DC Injection Current	<0.5% In				
Efficiency					
Max. Efficiency	97.6%				
Euro Efficiency	97.0%				
MPPT Efficiency	>99%				
Equipment Protection					
Integrated	DC Reverse Polarity Protection, AC Output Overcurrent Protection, Thermal Protection, AC Output Overvoltage Protection, AC Output Short Circuit Protection, DC Component Monitoring, Arc Fault Circuit Interrupter (AFCI)(optional), Anti-islanding Protection, DC Switch, Insulation Impedance Detection, Residual Current Detection				
Surge Protection Level	TYPE II(DC), TYPE II(AC)				
Interface					
LCD/LED Display	LCD				
Communication Interface	WIFI, RS485, CAN				
General Data					
Operating Temperature Range (°C)	-40 to +60°C, >45°C Derating				
Permissible Ambient Humidity	0-100%				
Permissible Altitude	3000m				
Noise (dB)	<60				
Ingress Protection(IP) Rating	IP 65				
Inverter Topology	Non-Isolated (solar), Isolated (battery)				
Over Voltage Category	OVC II(DC), OVC III(AC)				
Cabinet Size (WxHxD mm)	456×750×268.5 (Excluding Connectors and Brackets)				
Weight (kg)	51.9				
Type of Cooling	Intelligent Air Cooling				
Warranty	5 Years / Up to 10 Years* *Warranty period varies by installation country. Refer to the Warranty Policy for details.				
Grid Regulation	IEC 61727, IEC 62116, CEI 0-21, EN 50549, NRS 097, RD 140, UNE 217002, OVE-Richtlinie R25, G99, VDE-AR-N 4105				
Safety / EMC Standard	IEC/EN 61000-6-1/2/3/4, IEC/EN 62109-1, IEC/EN 62109-2				

MPHi-6KW#48VPVSE**Features:****Excellent Performance**

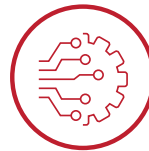
- 200% PV oversizing, 9kW Max. PV usable power
- Max. PV input current 36A
- Max. battery charge/discharge current of 140A
- Max. AC input power 11.5kW

**High Flexibility**

- Support 40-60V Lithium or Lead-acid battery
- Independent generator port
- Up to 12 pcs in parallel, 3-phase parallel system

**Safety & Reliability**

- IP54 Design
- 4 year Warranty
- Multiple built-in protections

**Intelligent Maintenance**

- USB OTG function
- Dual output with smart load
- Built-in Wi-Fi and bluetooth for local and remote monitoring



LEAVE POWER FOR MEDAL POWER

LEAVE POWER FOR MEDAL POWER

Technical Specifications

MPHi-6KW#48VPVSE		
Efficiency		
Efficiency (PV to AC)	95%	
Efficiency (BAT to AC)	93%	
PV Input		
Max. PV Usable Power	9,000W	
Max. PV Voltage	500V	
MPPT Voltage Range	60V-450V	
Star-up Voltage	60V	
Nos. of MPPT	1	
String Per MPPT	1	
Max. Input Current Per MPPT	36A	
Max. Short Circuit Current Per MPPT	45A	
Battery		
Battery Type	Lead-acid/Li-ion	
Rated Battery Voltage	48V	
Battery Voltage Range	40-60V	
Max. Solar Charging Current	140A	
Max. AC Charging Current	140A	
Max. Charging/Discharging Current	140A/140A	
AC Input		
Max. Input Power	11,500W	
Max. AC Input Current	50A	
Rated Grid Voltage	220V/230V/240V	
Input Voltage Range	170V-280V (For UPS)/90V-280V (For Home Appliances)	
Rated Grid Frequency	50Hz/60Hz	
AC Output		
Rated Output Power	6,000W/6,000VA	
Rate Output Voltage	220V/230V/240V	
Rated Output Current	27.3A/26A/25A	
Rate Output Frequency	50Hz/60Hz	
Max. Continuous AC Passthrough (grid to load)	40A	
Overload Capacitor	105%-150% after 10.5s shutdown / 150%-200% after 5.5s shutdown	
Surge Power	12000VA 5s	
THDv (@linear load)	< 3%	
Parallel Capacity	12 PCS	
Switch Time	10ms Typical (For UPS)/20ms Typical (For Home Appliances)/50ms Typical (For Parallel Operation)	
Protection		
Ingress Protection	IP54	
Over Voltage Protection	Yes	
Overload Protection	Yes	
Short Circuit Protection	Yes	
Over Temperature Protection	Yes	
Surge Protection	DC: Type III, AC: Type III	
General		
Communication	USB (OTG), RJ45 (for BMS,CAN/RS485), RJ45 (for Parallel), Dry-Contact, WI-FI/LAN (Optional)	
Operation Temperature Range	-10 ~ +55°C	
Humidity	5%-95%	
Max. Operating Altitude	4000m, >2000m derating	
Cooling Method	Intelligent fan cooling	
Display	LED+LCD	
Mounting	Wall-mounted	
Dimensions (W*H*D)	345mm*445mm*120mm	
Weight	10.8Kg	
Certification	EN/IEC62109-1, EN/IEC62109-2, EN/IEC 61000	
Warranty	4 Years	

Remarks: · The range of output voltage and frequency may vary depending on different grid codes.
· Specifications are subject to change without advanced notice.

Features:

- Accept extra high PV input current and power.
- Optional Dual output.
- Output power factor PF=1
- lithium battery activation start function with ac and solar mode
- Parallel function to expansion power
- Higher solar charging efficiency and battery protection
- Pure sine wave output, adaptable to different loads
- 500V high PV input
- Optional WIFI function
- Working with or without battery

BRHi-6.2KW#48VPVT



Technical Specifications

Model		BRHi-6.2KW#48VPVT
Input	Phase	L+N+PE
	Nominal Voltage	208/220/230/240VAC
	Voltage Range	154-264VAC±3V Normal Mode) 185-264VAC±3V UPS Mode
	Frequency Range	50Hz/60Hz(Adaptive)
Output	Rated Power	6200W
	Output Voltage(AC)	208/220/230/240VAC±5%
	On / Off grid	Yes
	Output Frequency	50/60Hz±0.1%
	Dual Output	Yes
	Waveform	Pure sine wave
	Transfer Time(Optional)	10ms for computer , 20ms for home appliances
	Peak power	12400VA
	Overload	Battery Mode 11s@105%~150% Load; 2s@150%~200% Load; 400ms@>200% Load.
	Peak efficiency (battery mode)	>94%
Grid feed	Output Voltage(AC)	220/230/240VAC
	Grid voltage	170-265VAC
	Grid frequency	49-51±1Hz/59-61±1Hz
	Output Current	26.9A
	Power factor	>0.99
	Transfer efficiency(DC/AC)	98%
Battery	Rated voltage	48Vdc
	Constant voltage charging voltage (Optional)	56.4Vdc
	Floating charge voltage (Optional)	54Vdc
charger	PV charging method	MPPT
	PV Input current	27A
	PV max input power	8500W
	MPPT tracking range	60~500Vdc
	Max PV input voltage	500Vdc
	Max PV charging current	120A
	Max AC charging current	100A
Display	Max charging current	120A
	LCD display	Operating mode/load/input/output, etc.
Communication port	RS232	5PIN/Pitch2.0mm, baud rate 2400
	Expansion & configuration port	2×5PIN/Pitch2.54mm, lithium battery BMS communication card, WIFI card, dry contact card, etc.
	BMS protocol	Pylon tech, Pace, Voltronic, Growatt, Deye, Leoch, Sofar, Goodwe, etc.
	Parallel interface	YES
Environmental parameters	IP Rate	IP21
	Operating environment temperature	-15~55℃
	Operating environment humidity	20%~95% Non condensing
	Storage temperature	-15~50
	Altitude	The altitude should not exceed 1000m, derating above 1000m, up to 4000m, refer to IEC62040
	noise	≤50db

FIDUS BATTERY PLUS



Superior Safety and Reliability

Multi-layer protection design, built-in fire suppression module



Designed for Extreme Conditions

Available in extreme cold region, against water jet, fully dustproofed, smoke & salty air resistant.



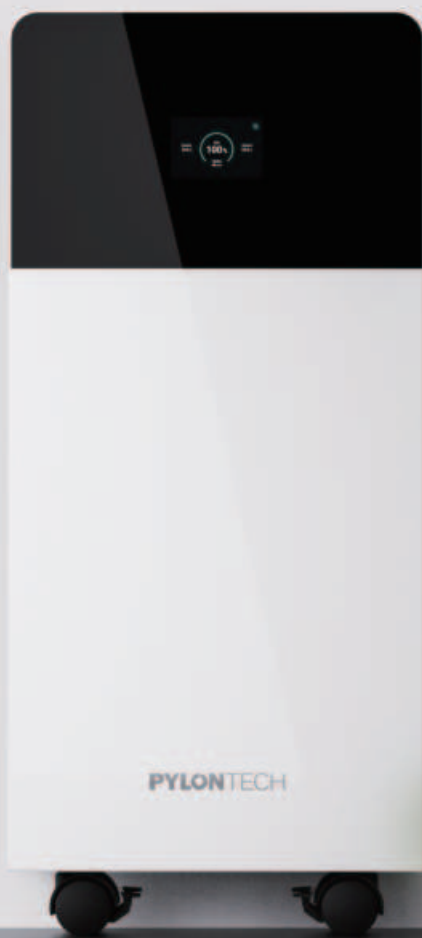
Flexible Installation

Wall-mounted or floor-standing.



Smart Monitoring

APP & LED display



YOUR HOME BATTERY


Fidus Battery Plus

Model

FB-L-16
FB-L-16-Pro

Nominal Voltage (Vdc)		51.2	
Nominal Capacity (Wh)		16076	
Usable Capacity (Wh)*		16076	
Depth of Discharge (%)*		100	
Dimensions (mm)		435(W)*240(D)*900(H)	
Weight (Kg)		130kg	
Discharge Voltage (Vdc)		40 ~ 56.8	
Charge Voltage (Vdc)		56 ~ 56.8	
Maximum Continuous Charge/Discharge Current (A)		200/200	
Peak Charge/Discharge Current (A)		300A@15s	
Communication		CAN, RS485	
Configuration (maximum quantity in one battery group)*		20	
Configuration (maximum strings)**		6	
Working Temperature (°C) ***	charging:	0~55	-20 ~ 55
	discharging:	-20~60	-20 ~ 55
Storage Temperature (°C)		-20 ~ 60	
Cooling Type		Natural	
Protective Class		I	
IP Rating of Enclosure		IP65	
Anti-corrosion		C5-M	
Humidity(%, RH, No Condensation)		5 ~ 95	
Altitude(m)		≤4000	
Certifications		IEC62619、IEC63056、UN38.3、EMC/CE	
Design Life (year) (25°C /77°F)		15	
Cycle Life (25°C /77°F) ÷*		8000	
Interaction		Display screen、Bluetooth and WIFI	
Fire extinguishing		Aerosol (Optional)	

*Battery supports 100% DoD; rated capacity at 25°C, 0.2C discharge. Default SOC lower limit set at 5%, adjustable.

** For multi-group setups, LV-HUB-V2 is required for parallel connection; acts as master. Max 19 batteries per group.

*** BMS limits charge/discharge current in extreme temps. Between -20°C and 0°C, battery heating activates to allow charging above 0°C.

**** Specs based on 25°C, 0.2C charge/discharge, SOC range 5%-95%.



Low Voltage Battery

LB-16D-G3

Description

The LB-16D-G3 is a deep cycle high capacity LFP battery with improved safety, long lifespan, and optimized user experience.

It has superior performance with a capacity of 16.08 kWh for a single unit and a maximum of 257 kWh for 16 batteries in parallel, which can meet the residential electricity demand.

Features

01

LFP with improved safety design

04

Adaptable to environments with temperatures ranging from -20°C to 55°C

02

Provides abundant power with 314 Ah capacity, ensuring long-lasting energy storage

05

Flexible capacity options from 16.08 kWh to 257 kWh

03

Supports wall-mounted, floor-standing, and stackable installation

06

Intelligent BMS with multiple protection functions

Technical Specifications

Model	LB-16D-G3
System Data	
Battery type	LiFePO ₄
Rated capacity (Ah)	314
Total energy (kWh)	16.08
Rated voltage (V)	51.2
Voltage range (V)	44.8-58.4
Max. continuous charging/discharging current (A)	160
Peak charging/discharging current (A)	240 (10s)
Communication	CAN
Recommended depth of discharge (DOD)	90%
Max. parallel quantity	16
Protection	
Overvoltage and undervoltage protection	Integrated
Overcurrent protection	Integrated
Overtemperature and undertemperature protection	Integrated
DC breaker	Integrated
Heating	Optional
General	
Dimensions (W × H × D [mm]) ⁽¹⁾	410 × 770 × 235
Weight (kg) ⁽¹⁾	110
Installation Environment	Indoor/Outdoor
Mounting	Wall mounted/Floor standing/Stackable
Charging temperature (°C)	0 to +55
Discharging temperature (°C)	-20 to +55
Protection degree	IP65
Cooling	Natural convection
Altitude (m)	≤2000
Cycle life (25°C)	8000 Cycles
Certification	IEC 62619, UN 38.3, CE-EMC, CB
Warranty ⁽²⁾	5-10 Years

(1) The actual dimensions and weight may differ. For details, please contact Hoymiles sales.

(2) Please refer to Hoymiles Warranty Terms & Conditions for more details.

RESIDENTIAL BATTERY

MANA 5.1 / 10.4 / 16.0 / 20.0

Low Voltage

UN38.3



MANA 5.1

MANA 10.4

MANA 16.0

MANA 20.0



Features



One - Touch Parallel
Scaling



Remote App Control
& Updates



Mainstream Inverter
Compatibility



Smart BMS & Precise
Detection and Protection



Auto Aerosol Fire Extinguishing
Ultimate Security, Strengthen Defenses



Scenarios



Houses



Off-grid Cabins



Small Stores



Small Offices



Small Farms

DATASHEET

Model	MANA 5.1	MANA 10.4	MANA 16.0	MANA 20.0
Performance				
Cell technology	LFP (LiFePO ₄)			
Battery usable energy [1]	5.12 kWh	10.49 kWh	16.07 kWh	20.07 kWh
Nominal capacity	100 Ah	205 Ah	314 Ah	392 Ah
Nominal voltage	51.2 V			
Operating voltage	44.8 ~ 56.16 V			
Max. charge and discharge current [2]	100 A	200 A	200 A	200 A
Recommend Depth of Discharge	90%			
Communication				
Display	SOC status indicator, LED indicator, LCD display (Optional)			
Communication	CAN / RS485 / RS232 / Wi-Fi			
General Specification				
Dimension (W×D×H)	390×160×643 mm	420×245×706 mm	485×245×785 mm	500×270×892 mm
	15.35×6.3×25.31 inch	16.73×9.65×25.31 inch	19.1×9.6×30.91 inch	19.7×10.6×35.1 inch
Weight	47 Kg (103.6ls)	82 kg (180.77 lbs)	122 kg (265.96 lbs)	145 kg (319.7 lbs)
Installation	Floor stand or Wall mounted			
Operating temperature [3]	Charge: 0 to 50°C (32 to 122°F) Discharge: -15 to 50°C (5 to 122°F)			
Environmental humidity	≤ 95%RH (No condensation)			
Ingress protection rating	IP20			
Warranty Period [4]	5 years / 10years (Optional)			
Scalability	Max. 15 batteries in parallel			
Application	ON Grid / ON Grid + Backup / OFF grid			
Compatible inverters	Refer to compatible inverter list (Compatible with major PCS brands)			
Standard Compliance				
Compliance	UN38.3 / IEC62619 / IEC61000 / IEC62040-1 (More available upon request)			

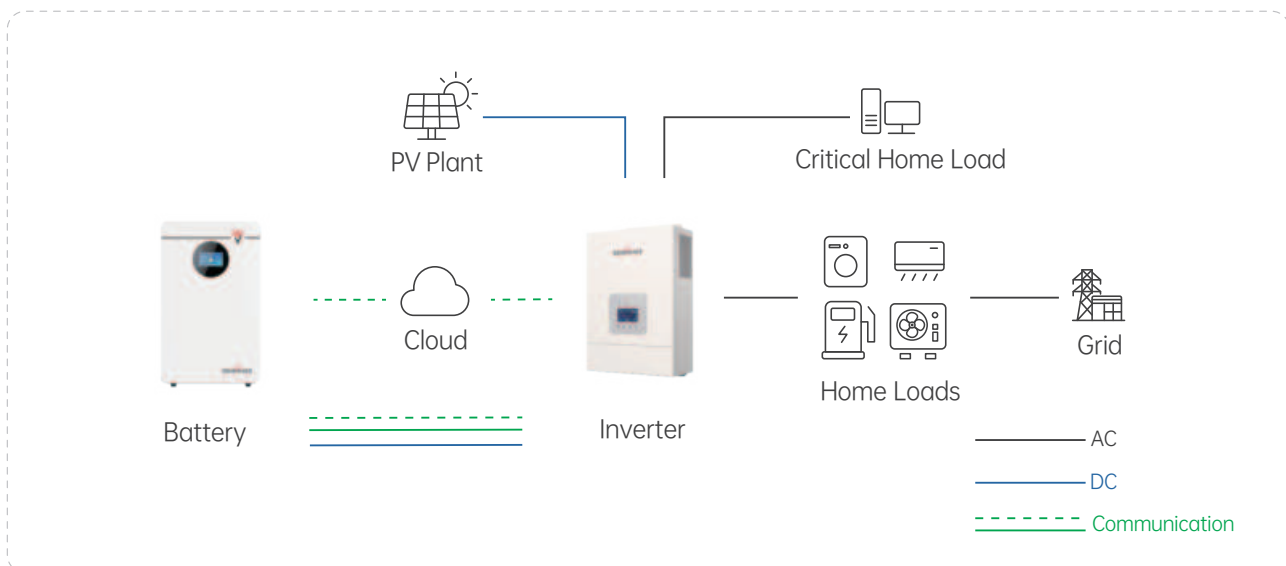
[1] Test conditions: 100% depth of discharge (DOD), 0.2C rate charge & discharge at 25°C.

[2] There is 0.5C or 1C configurations optional in factory default.

[3] Charge/discharge derating occurs when the temperature is below 0°C or above 45°C.

[4] Please refer to the Warranty Letter for applicable conditions.

SYSTEM DIAGRAM



LITHIUM BATTERY

LifePO4

Features

- This series is a complete solution for the lithium iron phosphate battery system for the home energy storage field. There are three types of wall-mounted, vertical structures and rack. The system is safe and reliable. It can be used in home storage, industrial and commercial energy storage and other fields

51.2V-100Ah



Product Features



Safe, reliable
and long life



Dynamic identification, automatic
paralleling, no need for DIP switches



Support Bluetooth, mobile
APP switching PCS protocol



Flexible expansion, the number of
parallel machines supports up to 15



CAN/RS485/
LAN interface



Supports mixed parallel
of old and new batteries



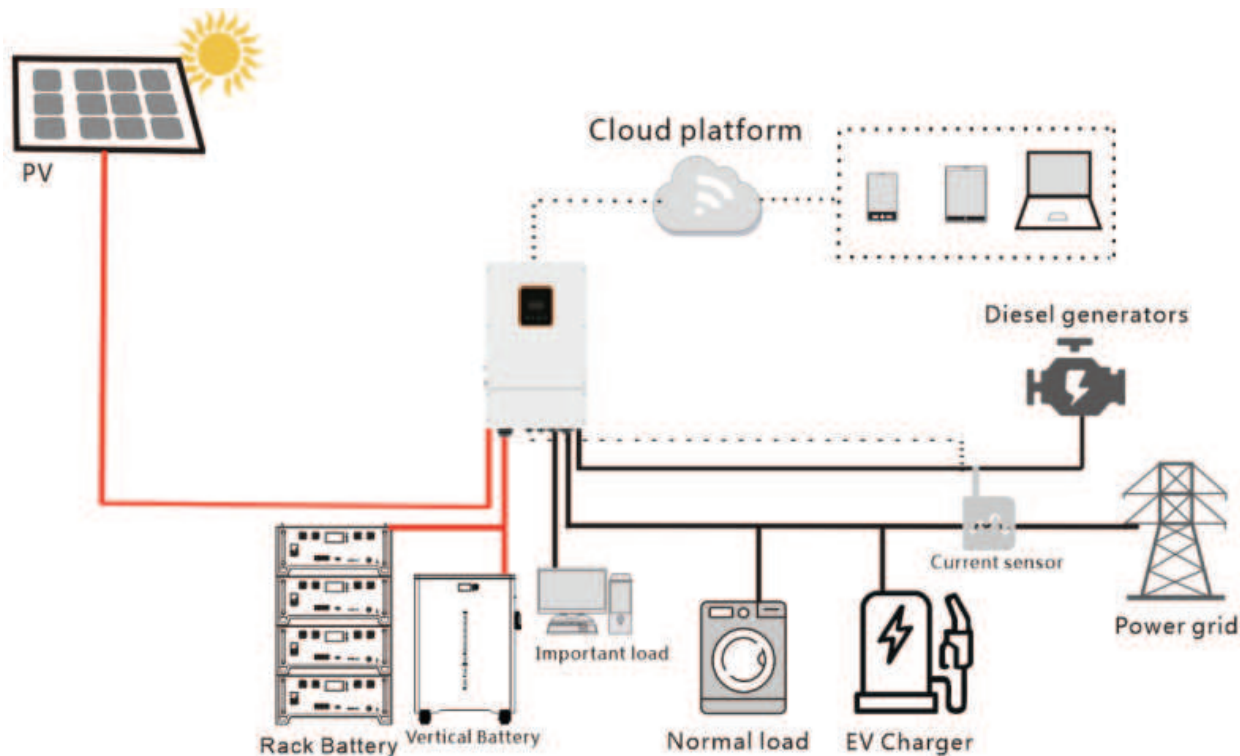
Efficient operation
and maintenance

Compatible With Inverter Brands

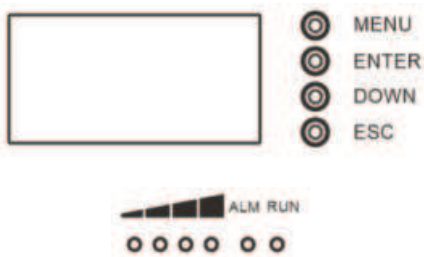
MACQUARIE	Solis	GROWATT	DEYE	VITVON	SERMATEC	LUXPOWER	GSSTES
SMA	GOODWE	SREN	HuiNeng	iYPower	OUKITEL	ROBOROCK	Sunsunk
Voltronic	Pylontech	MustPower	SAKO	Senergy	Felicity	KSTAR	SOROTEC

LITHIUM BATTERY

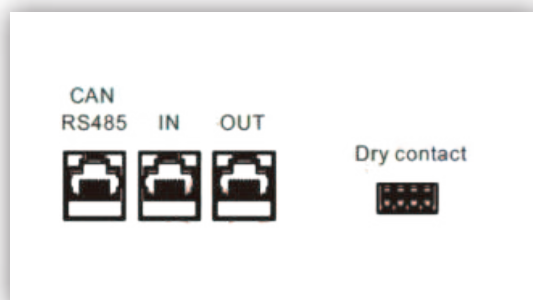
LifePO4



System Specifications



Panel silkscreen preview



Battery communication interface preview

Model	WT-5121000-LT
Voltage(V)	43.2~58.4V
Cell	3.2V100Ah
Number of parallel machines	15PCS
Nominal voltage(V)	51.2
Nominal Capacity(Ah)	100
Nominal Energy(kWh)	5.12
Maximum power output(kWh)	3.84
Maximum discharge current(A)	75
Maximum charging current(A)	50
Cut-off voltage(V)	48(40)
Maximum charge voltage(V)	56(58.4)
Cycles(25°C)	≥4000
Charging temperature(°C)	0~55
Discharge temperature(°C)	-20~55
Operating humidity	<95%R.H
Communication mode	CAN/RS485
Optional modules	Bluetooth
Product size(W*D*H,mm)	480*442*155
Packing size(W*D*H,mm)	565*515*255
Protection class	IP21
System weight(kg)	Approx 48